

UNIT

27

BIOLOGY & HUMAN WELFARE





Human Welfare and Disease Management

Key Concepts:

- **Human welfare** involves biological applications that benefit human health, agriculture, and society. It includes the improvement of plants and animals, food production, medical advancements, and disease management.

Major Topics in Human Welfare:

1. **Vaccination and Integrated Disease Management**
2. **Animal Husbandry**
3. **Techniques to Enhance Crop and Fruit Yield**
4. **Home Gardening**
5. **Role of Microbes in Human Welfare**

Vaccination and Integrated Disease Management:

1. **Vaccination:**

- **Definition:** The process of administering a vaccine to develop artificial immunity against infectious diseases.
- **Purpose :** To stimulate the body's immune system to recognize and fight off specific pathogens without causing the disease.

2. **Integrated Disease Management (IDM):**

- **Definition:** The coordinated use of multiple measures to control diseases in populations, including preventive, protective, and resistance strategies.
- **Approaches:**
 - **Control :** Managing the pathogen directly.
 - **Protection:** Preventing disease spread through measures like vaccines.
 - **Resistance:** Promoting immunity in populations.
 - **Prevention:** Avoiding environmental conditions that promote disease.

3. **Key Principles of Disease Management:**

- **Primary Prevention :** Preventing diseases before they occur (e.g., vaccination, healthy living).
- **Secondary Prevention:** Early detection and treatment of diseases (e.g., screening programs).
- **Tertiary Prevention:** Minimizing disease impact and recurrence (e.g., rehabilitation and lifestyle adjustments).

Integrated Disease Management Strategies:

1. **Disease Control:**

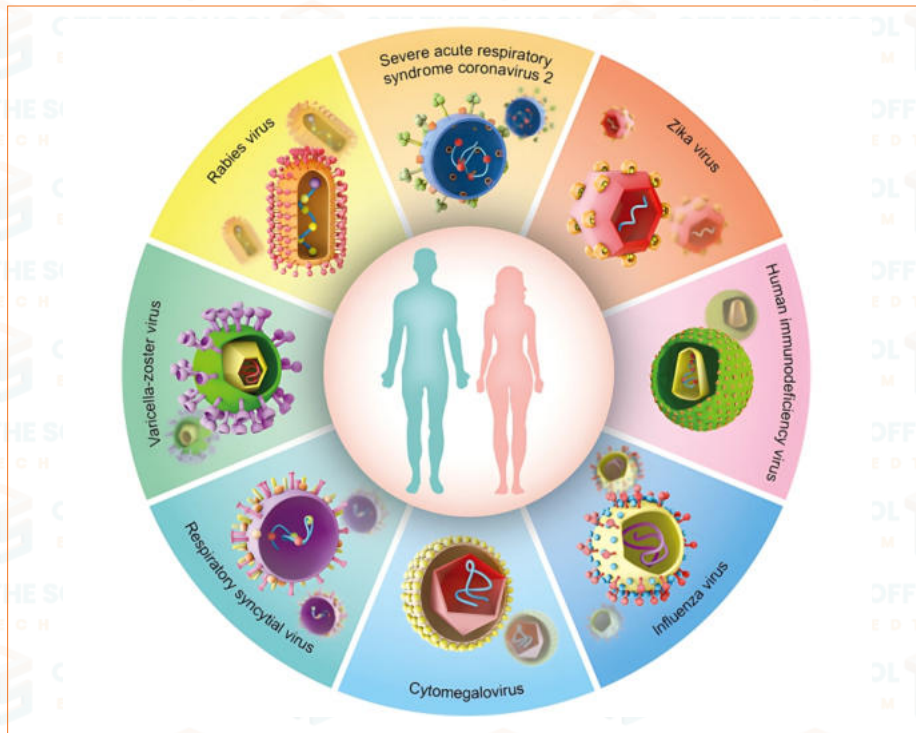
- Reactive measures to address existing diseases.
- Involves identifying pathogens and treating affected individuals or communities.





2. Preventive Measures:

- **Environmental Control:** Modifying environments to reduce pathogen transmission (e.g., proper sanitation, safe water).
- **Personal Hygiene:**
 - **Handwashing:** Proper handwashing with soap and water is essential in preventing disease transmission.
 - **Food Handling:** Ensuring hygiene when preparing food to avoid contamination.
- **Respiratory Hygiene:** Using tissue papers or handkerchiefs while coughing and sneezing to prevent the spread of infections.
- **Avoiding Zoonotic Diseases:** Limiting direct contact with wild animals to prevent diseases that can transfer from animals to humans.
- **Avoiding Sharing Personal Items:** Reduces the risk of disease spread (e.g., not sharing personal hygiene items).



Impact on Human Welfare:

Effective disease management and vaccination contribute significantly to improving public health, preventing outbreaks, and reducing the burden of infectious diseases on societies.





Major Mechanisms of Disease Management

Major Mechanisms of Disease Management

1. Host Resistance Against Causative Agents

- Resistance found in plants and animals helps limit or block disease development.
- Includes:
 - Physical barriers (e.g., skin, cell walls).
 - Rapid immune responses that neutralize or kill pathogens.

2. Biological Control

- Uses living organisms (predators, parasites, or competitors) to reduce disease severity.
- Example: Introducing predator species to control pest populations.

3. Cultural Control

- Involves environmental and agricultural practices to reduce disease.
- Techniques include:
 - Intercropping.
 - Crop rotation.
 - Field sanitation.
 - Adjusting planting/spread dates.
- Often small individual effects, but powerful when combined with other methods.

4. Chemical Control

- Uses chemicals like pesticides, insecticides, fungicides, and bactericides to control diseases.
- Applied especially in vector-prone areas.
- **Drawbacks:**
 - Human poisoning risks.
 - Destruction of beneficial insects.
 - Crop damage.
 - Residual chemical buildup.

Aims and Objectives of Integrated Disease Management (IDM)

- **Main goals:**
 - Stop disease spread.
 - Test and apply effective preventive measures.
 - Remove or reduce biological causes of diseases.
 - Engage community participation and raise public awareness (media, workshops, seminars).
 - Promote proactive actions (prevention) rather than reactive responses.
 - Emphasize safe practices like seeking professional medical help instead of self-medication.





Vaccination and Its Importance

What is a Vaccine?

- A suspension containing weakened or killed microorganisms designed to stimulate the immune system.
- Administered via:
 - Injection.
 - Oral intake.
 - Nasal sprays.



Types of Vaccines

- **Inactivated** (killed microorganisms).
- **Live attenuated** (weakened microorganisms).
- Others include protein-based, toxin-based, mRNA vaccines.

Historical Milestones

- **Edward Jenner (1796)**: Developed the first vaccine using cowpox to protect against smallpox.
- **Louis Pasteur (1881)**: Developed immunization against anthrax using attenuated bacillus.

How Vaccines Work?

- Train the immune system to recognize and fight specific pathogens.
- Reduce disease transmission in the community.
- Protect both vaccinated individuals and those around them.

Importance of Vaccination

- Prevents severe diseases and fatalities.
- Reduces global illness burden (prevents ~3 million deaths/year).
- Protects against:



- Pneumococcal disease.
- Influenza.
- HPV.
- Hepatitis B.
- COVID-19.
- Whooping cough.
- Chickenpox (Varicella vaccine).
- Tetanus (Tdap, Td vaccines).

Vaccine Safety

- All vaccines undergo strict testing before approval.
- Mild side effects (fever, swelling, tenderness) are common but temporary.
- Overall, the **benefits far outweigh the risks**.

List of Some Viral Diseases and the Role of Vaccines

Table — List of Some Viral Diseases

VIRUS	DISEASE	SYMPTOMS	VACCINE
Coronavirus	Common cold, COVID-19	Fever, headache, fatigue, chest pain, shortness of breath	Sinopharm, Sinovac, Pfizer, Moderna, AstraZeneca
Pox viruses	Cowpox, Smallpox	Fever, sores, rashes	Varicella vaccine
Polio virus	Poliomyelitis (Polio)	Fever, loss of appetite, nausea, sore throat, malaise, constipation, abdominal pain	Inactivated Polio Vaccine (IPV), Oral Polio Vaccine (OPV)
Hepatitis viruses	Hepatitis (A, B, C, D, E)	Fever, muscle aches, loss of appetite, nausea, vomiting, fatigue, jaundice	Hepatitis B vaccine (for A & B)
Measles, mumps, rubella viruses	Measles, mumps, rubella	High fever, tiredness, cough, red eyes, runny nose, red rash starting at head and spreading down	MMR vaccine (Measles, Mumps, Rubella)

The Role of Vaccines in Preventing Polio, Measles, Hepatitis, and Tetanus

Vaccines **play a vital role** in preventing serious infectious diseases like **polio, measles, hepatitis, and tetanus**, protecting individuals and communities, reducing disease transmission, and preventing complications or death.



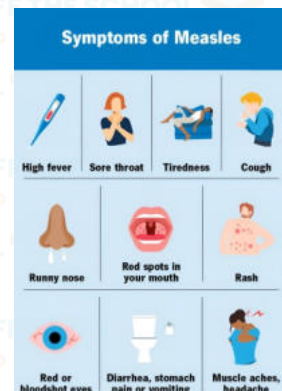
i) Vaccination Against Polio

- **Disease:** Poliomyelitis (Polio), caused by polio virus.
- **Transmission:**
 - Person-to-person.
 - Through fecal contamination, saliva, contaminated food/water.
- **Symptoms:**
 - Fever, sore throat, nausea, vomiting, abdominal pain, constipation.
- **Vaccines:**
 - **IPV (Inactivated Polio Vaccine)** — given by injection.
 - **OPV (Oral Polio Vaccine)** — given orally, mainly for children under 5.



ii) Vaccination Against Measles

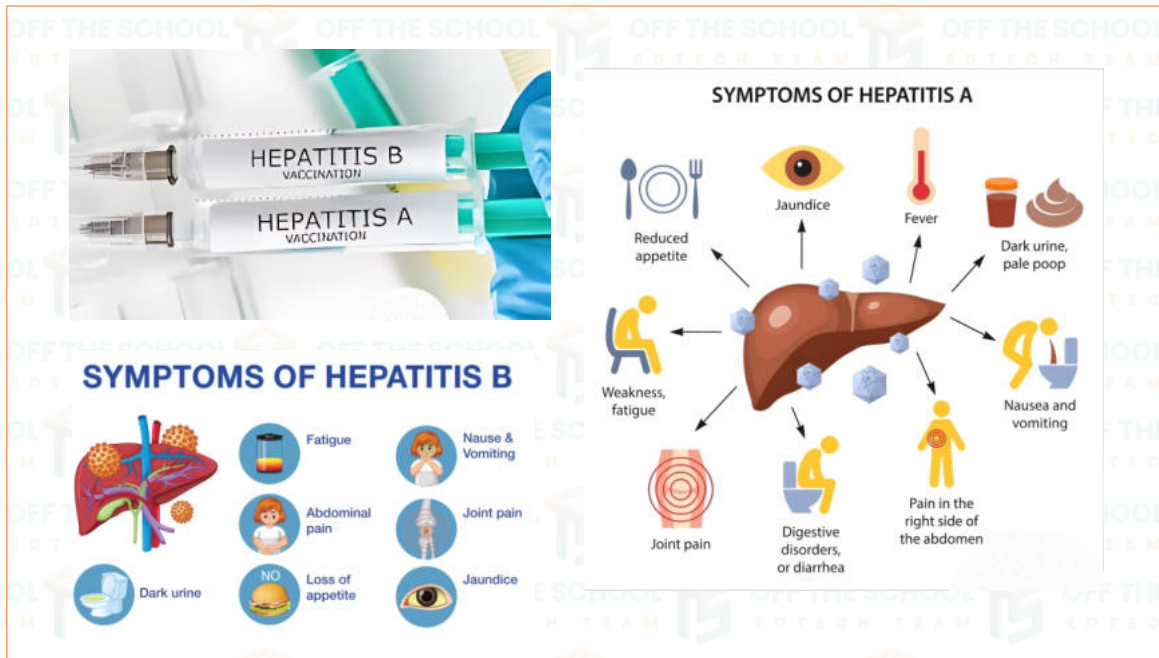
- **Disease:** Measles, caused by a paramyxovirus.
- **Transmission:**
 - Airborne (respiratory droplets, aerosol particles from coughs/sneezes).
- **Symptoms:**
 - High fever, cough, red eyes, runny nose, rash starting on head and spreading down.
- **Vaccine:**
 - **MMR Vaccine (Measles, Mumps, Rubella)** — contains live attenuated viruses, highly effective and safe.





iii) Vaccination Against Hepatitis

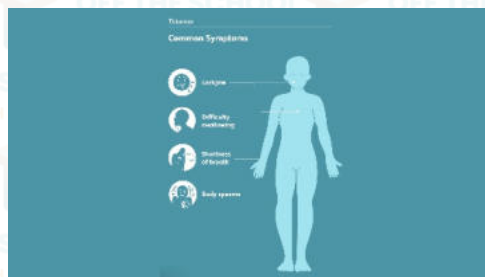
- **Disease:** Hepatitis (inflammation of liver), caused by hepatitis viruses (A, B, C, D, E).
- **Transmission:**
 - Hepatitis A/E → contaminated food and water.
 - Hepatitis B/C → body fluids, blood.
 - Hepatitis D → blood contact.
- **Symptoms:**
 - Flu- like symptoms, abdominal pain, jaundice (yellowing of skin/eyes, dark urine).
- **Vaccines:**
 - **Hepatitis A vaccine** — inactivated virus.
 - **Hepatitis B vaccine** — recombinant DNA technology.
 - **Note:** No vaccine currently for hepatitis C.



iv) Vaccination Against Tetanus

- **Disease:** Tetanus ("lockjaw"), caused by *Clostridium tetani* bacteria (produces neurotoxin).
- **Transmission:**
 - Entry through wounds, cuts contaminated with soil or dust.
- **Symptoms:**
 - Muscle stiffness, painful spasms, lockjaw, breathing difficulties, feeding problems, constipation.
- **Vaccines:**
 - **DT vaccine** — Diphtheria + Tetanus.
 - **Tdap vaccine** — Tetanus, Diphtheria, Pertussis (whooping cough).





Summary Importance

- Vaccines **train the immune system** to recognize and fight infections.
- Widespread vaccination has controlled or eliminated diseases in many regions.
- Protects **both individuals and communities** from severe illness, complications, and death.

Schedule of Vaccination & Animal Husbandry

Table — Schedule of Vaccination

DISEASE	VACCINE	AGE GROUP
Polio	OPV (Oral Polio Vaccine), IPV (Inactivated Polio Vaccine)	Birth to 5 years
Measles	MMR Vaccine (Measles, Mumps, Rubella)	Up to 1 year
Hepatitis A	Hepatitis A Vaccine (inactivated)	At any age
Hepatitis B	Hepatitis B Vaccine (recombinant DNA)	At any age
Tetanus & Diphtheria	DT Vaccine (Tetanus & Diphtheria), TDP Vaccine (Tetanus, Diphtheria, Pertussis/Whooping cough)	Childhood

Animal Husbandry

Definition:

Animal husbandry is the branch of agriculture dealing with the management, care, and breeding of livestock (buffaloes, cows, cattle, sheep, goats, horses, poultry) for producing milk, meat, eggs, fiber, and other food products.





Key Aspects of Animal Husbandry

- Providing proper **shelter, feeding, water, hygiene, and medical care**
- Managing **breeding, calving (birthing), offspring care**, and shade/weather protection.
- Access to **veterinary care** and ensuring overall animal welfare.

Importance

- Provides highly nutritious food products (meat, milk, eggs).
- Supports commercial markets and household needs.
- Includes **dairy, poultry, apiculture (beekeeping), aquaculture (fish farming)**.

Major Types of Animal Husbandry

Dairy Farming

- Focus: Milk production for dairy products (cheese, yogurt, butter).
- Involves cows, buffaloes, sheep, goats.
- Requires regular animal care, physical & mental well-being checks.



Poultry Farming

- Focus: Breeding and raising hens, ducks, turkeys for meat & eggs.
- Important for both commercial supply and household consumption.
- Demands disease-free, healthy environments.





Apiculture (Bee Farming)

- Focus: Managing bee colonies in hives.
- Provides honey, beeswax, and supports pollination.
- The place for keeping bees is called an **apiary**.



Role of Livestock in National Economy

- Livestock contributes **~11.9% to Pakistan's GDP** (Economic Survey 2021–2022)
- Provides:
 - Milk, meat, eggs, wool, by-products.
 - Food security, nutrition, income for rural and urban populations.
 - Draught power (for farming & transport) and organic fertilizer.

Employment: Engages ~37.4% of the labor force.

Growth Factors:

- High yields, favorable prices.
- Government support, better access to resources.

Pakistan's Global Status

- Leading producer of milk and meat.

Challenges & Opportunities

- Low productivity.
- Poor feed quality.
- Weak healthcare services and infrastructure.
- Environmental issues.

Opportunities:

- Adoption of modern technologies.
- Improvement in animal welfare and processing.
- Strengthening export potential and market competitiveness.





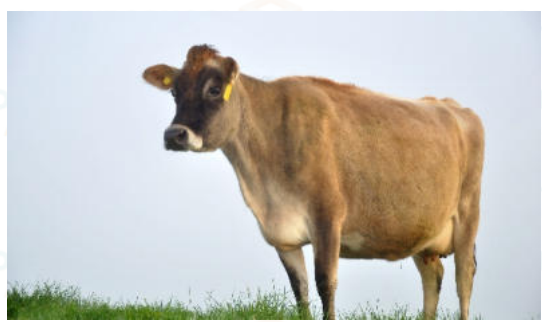
Outstanding Milk Producing Breeds & Other Topics

Outstanding Milk Producer Breeds of Cows in Pakistan

Pakistan’s second-leading milk producers after buffalo are cows.

- There are **five main outstanding cow breeds**:

COWS BREED	PER LACTATION CYCLE (LITERS)
Brown Swiss	10,000–12,000 L (Best milk producer)
Simmental	~9,000 L (Best milk producer)
Normande	6,000–7,000 L
Jersey	5,000–6,000 L
Holstein	4,000–5,000 L





Note: A lactation cycle is the period between one calving (birth) and the next, covering early, mid, late lactation, and a dry period.

Outstanding Milk Producing Breeds of Buffalo

Buffalo is the **chief milk-producing species** in Pakistan.

BUFFALO BREED	PER LACTATION CYCLE (LITERS)
Nili-Ravi	1,500–2,500 L (Top producer)
Kundi	1,700–2,000 L (Second best in Pakistan)
Jaffrabadi	1,000–1,200 L
Godavari	1,200–1,500 L
Bhadawari	800–1,000 L





Latest Techniques to Enhance Crop & Fruit Yield

Plant Breeding:

- Improves plant traits and increases biodiversity.
- Mendel is called the **Father of Plant Breeding**.

Methods of Plant Improvement

1. Acclimatization

- Plants naturally adjust to environmental changes (temperature, salinity, humidity).
- **Plant introduction**: Transferring plants to new areas (e.g., papaya from Mexico, potatoes from Peru).

2. Selection

- Choosing the best plants from a crop population to propagate desirable traits.

3. Hybridization

- Crossing two genetically dissimilar parents.
- Produces hybrids showing **heterosis** (hybrid vigor), better yield, disease resistance.
- Used in crops like onions, tomatoes, oilseeds, pulses, fruits.

4. Backcrossing

- Crossing a hybrid back to one parent or a similar plant.
- Used to introduce specific traits (denoted as BC₁, BC₂ generations).

Home Gardening

- Combines farming and gardening around the home.
- Involves vegetables, fruits, flowers, ornamental plants.





Importance of Home Gardening

- **Nutritional Value:** Provides fresh, organic produce.
- **Health Benefits:** Reduces stress, promotes exercise.
- **Environmental Effects:** Plants absorb CO₂, release O₂, reduce erosion.
- **Budget Control:** Reduces market expenses, year-round food availability.
- **Social Benefits:** Space for family activities, beautifies the home

Seasonal Vegetables & Fruits for Home Gardening

SUMMER VEGETABLES	SUMMER FRUITS
Green beans, French beans	Mango, apricots
Squash, carrot, cauliflower	Plums, leeches, grapes
Garlic, onion, okra	Guava, papaya, peaches
Brinjal, cabbage, chilies	Watermelon, grewia
WINTER VEGETABLES	WINTER FRUITS
Asparagus, peas, garlic	Oranges, banana
Spinach, potato, tomato	Grapefruit, date
Broad beans, onion, lettuce	Pummelo, kiwifruit
Spring cabbage	Passion fruit, pear, tangerines, pomegranate

Role of Microbes in Human Welfare

- Microbes include bacteria, viruses, parasites; found in water, soil, air.
- Play roles in antibiotics, vaccines, enzymes, vitamins, and food fermentation.

Beneficial Viruses

- **Cowpox virus** → Immunity against smallpox.
- **Oncolytic viruses** → Kill cancer cells, support immunity.
- **Bacteriophages (phages)** → Kill bacteria.
- **Endogenous retroviruses** → Helped placenta evolution.
- **Gamma herpes virus (MHV-68)** → Provides infection resistance.

Microbes in Household Food Processing

FOOD	MICROBE & ROLE
Bread	Yeast (<i>Saccharomyces cerevisiae</i>) — produces CO ₂ , makes dough rise, gives soft texture.
Yogurt	<i>Lactobacillus</i> species — ferments lactose into lactic acid, gives tangy taste, thickness.
Cheese	<i>Propionibacterium shermanii</i> (Swiss cheese), <i>Penicillium roqueforti</i> (Roquefort cheese) — create flavor, texture.





Role of Microbes in Industrial Production, Sewage Treatment & Energy Generation

1. Role of Microbes in Industrial Production

- Microbes play a key role in **industrial and food production**, contributing to the manufacture of:
 - Chemicals
 - Hormones
 - Antibodies
 - Beverages
 - Fruit juices
 - Enzymes
- Fermented Beverages**
 - Yeast (*Saccharomyces cerevisiae*)** → same yeast used in bread-making.
 - Converts sugars into **alcohol** during fermentation → produces alcoholic drinks like **beer** and **wine**.

2. Role of Microbes in Sewage Treatment

- Sewage treatment removes pollutants from wastewater to reduce environmental harm.
- Microbes, especially **Bacillus species**, help break down proteins, oils, and organic materials.

Main Stages of Sewage Treatment

STAGE	DESCRIPTION
Primary Treatment	Removal of floating solids & heavy organic matter. Sewage flows through a sedimentation tank , where solids settle at the bottom.
Secondary Treatment	Microbes digest residual organic matter through aerobic or anaerobic processes, reducing organic load. Microbes grow & multiply while breaking down materials.
Sludge Digestion	Biological process decomposing organic solids into stable, reduced volume substances, making sludge easier to manage.
Tertiary Treatment	Final purification of treated water via filtration, disinfection, nutrient removal , ensuring water quality before reuse or discharge.
Effluent	The cleaned, treated wastewater that is released or reused after treatment.

3. Role of Microbes in Energy Generation

Microbes are vital for **producing biofuels and energy sources** such as:

- Hydrogen
- Methane
- Lipids
- Ethanol